FLIGHT DELAY AND AIRPORT OPERATIONS

Complete project presentation dialogue

From project objective and source data through Bronze, Silver, Gold, orchestration, validation, machine learning and dashboard

Suggested speaking time: approximately 15–20 minutes at a steady pace, depending on screen navigation. If your slot is strictly 10–15 minutes, shorten the source-data and layer descriptions.

1. Opening – Project Overview

“Good morning/afternoon everyone. Today, we are presenting our project, **Flight Delay and Airport Operations**.

The main objective of this project is to bring together flight schedules, flight movement events, weather observations and reference data, process them through a structured data pipeline, and make the resulting information available for operational analysis and reporting.

We want to make it easier to understand flight volumes, cancellations, airport and airline performance, fleet information, and the relationship between recorded delays and weather observations.

Our solution uses Azure storage and Azure SQL as source systems, Databricks for data processing and orchestration, Delta tables organized through the Medallion Architecture, and Databricks AI/BI dashboards for reporting.

At a high level, our data moves through three layers: Bronze for raw ingested data, Silver for cleaned and validated data, and Gold for business-ready analytical tables. We also have workflow orchestration, pipeline validation and an experimental machine-learning notebook.”

2. Source Data and Data Understanding

“We worked with six datasets. Three are operational datasets and three are reference or lookup datasets.

Operational datasets

• **flight_schedules:** contains planned flight information.

• **flight_movement_events:** records events such as departure, arrival, gate changes and boarding.

• **weather_observations:** contains weather measurements associated with airports and observation times.

Reference datasets

• **aircraft_master:** provides aircraft information.

• **airline_master:** provides airline information.

• **airport_master:** provides airport information.

“The distinction is that operational data records activities or observations, while lookup data describes entities such as airlines, airports and aircraft. The reference data helps us enrich and interpret the operational records.”

“Across the six datasets, we had 12,482 source rows: 11,171 flight schedule rows, 603 movement-event rows, 297 weather observations, 146 aircraft records, 137 airline records and 128 airport records.”

“These datasets form the input to our pipeline. We use the reference data to add context to operational records and to support the analytical model later in the Gold Layer.”

3. Bronze Layer – Raw Data Ingestion

“The Bronze Layer is our raw ingestion layer. Its purpose is to bring source data into Databricks while preserving the original information and adding ingestion context, rather than applying extensive business transformations at this stage.”

“For files stored in Azure Data Lake Storage Gen2, we use **Auto Loader**. Auto Loader incrementally discovers new files and ingests them into Bronze Delta tables. It maintains checkpoint information so it can track which files have already been processed.”

“For reference data coming from Azure SQL, we use a JDBC-based ingestion approach to bring the master tables into Bronze.”

“We use Auto Loader with **trigger(availableNow=True)**. This means it processes the files that are currently available and then stops. It uses Structured Streaming internally, but our implementation is batch-oriented incremental ingestion, not a continuously running real-time stream.”

“The main benefit of Bronze is traceability. We preserve source information so that if a data-quality issue appears later, we can investigate the original record and its ingestion context.”

4. Silver Layer – Cleaning and Data Quality

“After ingestion, the data moves into the Silver Layer. This is where we clean, validate and standardize the source records before using them for analytics.”

“Our Silver notebook defines business keys, required fields and data types for each dataset. It checks that the expected columns exist, trims text fields, converts columns into the required data types and checks mandatory fields.”

“It also detects failed type conversions, missing required values and invalid schedule chronology, such as an arrival timestamp that is earlier than or equal to the departure timestamp. Records that fail these checks are separated into valid and rejected datasets, with a reason recorded for rejection.”

Conflicting flight schedules

“One important finding was that 37 flight IDs had conflicting schedule records, involving 74 rows. Because we did not have a reliable effective-time field to determine which schedule version was correct, we quarantined all 74 records rather than arbitrarily choosing one.”

Deduplication

“We apply different deduplication logic depending on the dataset. For schedules, we remove exact duplicate business records rather than treating every repeated flight ID as a duplicate. For other datasets, we use their configured business keys and deterministic ordering rules to select the retained record.”

Referential integrity

“We also check relationships between operational data and reference data. For example, a schedule's airline, aircraft and origin and destination airports should exist in their respective master tables. We use orphan checks to identify records whose references cannot be found. These counts are reported for review rather than silently deleting the records.”

“Finally, the cleaned datasets are written as Silver Delta tables. The Bronze data remains unchanged. The Silver layer also retains rejection audit information so that rejected records can be reviewed.”

5. Gold Layer – Business-Ready Data Model

“The Gold Layer takes the cleaned Silver data and organizes it for business reporting. Our notebook creates nine Gold tables: four dimensions, four fact tables and one daily KPI mart.”

Dimension tables

• **dim_aircraft:** descriptive aircraft information.

• **dim_airline:** descriptive airline information.

• **dim_airport:** descriptive airport information.

• **dim_date:** date attributes used for time-based analysis.

“The dimension tables provide descriptive context. The fact tables store measurable flight and operational records.”

Fact tables

• **fact_flight_schedule:** planned flight schedule details.

• **fact_flight_activity:** flight activity and status information, including actual departure and delay-related fields where available.

• **fact_flight_movement:** movement events such as departure and arrival.

• **fact_weather_observation:** weather observations associated with airports and times.

“The Gold notebook connects facts to dimensions using keys. It also identifies actual departure information from relevant movement events and calculates departure delay by comparing actual departure time with scheduled departure time where the necessary timestamps are available.”

Daily KPI mart

“The ninth table is **mart_flight_daily**. It aggregates daily flight metrics by flight date and origin airport. It includes measures such as total flights, cancellations, departures, average departure delay, on-time departures, cancellation rate and punctuality percentage. This mart contains 5,604 rows.”

“The Gold tables are rebuilt using overwrite logic when the notebook runs. This gives us a repeatable snapshot-style build rather than an incremental CDC merge process.”

6. Databricks Job – Pipeline Orchestration

“To automate the processing stages, we created a Databricks Job named **capstone-03-flight_delay**. Instead of manually executing each notebook, the Job runs the tasks in a defined order and manages their dependencies.”

“Our current workflow has six tasks:”

• **bronze_adls_autoloader:** ingests the ADLS files into Bronze.

• **bronze_azure_sql:** ingests the Azure SQL reference data.

• **data_profile:** profiles the ingested data.

• **silver_transform:** cleans and validates the data.

• **gold_build:** builds the dimensions, facts and KPI mart.

• **validate:** runs the final pipeline validation notebook.

“The two Bronze ingestion tasks are independent and can run in parallel. The profiling task waits for both. Silver runs after profiling, Gold runs after Silver, and final validation runs after Gold.”

“This dependency structure is a DAG, or Directed Acyclic Graph. It represents the tasks and the order in which they are allowed to execute.”

File Arrival trigger

“We configured a File Arrival trigger on the Job using a Unity Catalog Volume location. When a new file arrives at the configured location, the trigger can start the workflow. The trigger starts the Job; Auto Loader is the component that discovers and ingests the files.”

“We also configured retries, with a two-minute delay and up to three retries, and notifications for Job start, success and failure. We can use the Job run history to see task outcomes and investigate failures.”

Technical challenge – storage permissions

"One technical challenge was configuring the File Arrival trigger because Databricks needed appropriate access to monitor the configured storage location. An Azure role commonly used for data access in this architecture is **Storage Blob Data Contributor**, assigned to the Databricks Access Connector managed identity at the relevant storage scope. The exact role assignment should be confirmed from Azure IAM before describing it as the role we granted. This Azure storage role is different from Unity Catalog privileges such as USE CATALOG, USE SCHEMA and READ VOLUME."

"Another important distinction is that manual notebook execution can work while an automated trigger still has an access or configuration issue. We therefore had to consider the permissions and identity used by the automated workflow, not just whether the notebook ran manually."

7. Pipeline Validation Notebook

"After the Gold build, we run a separate validation notebook. Its purpose is to check whether the pipeline outputs are present and consistent."

"The notebook uses helper functions. **count_rows()** counts records in a table, and **add_result()** records each validation outcome as a pass or fail."

Main validation checks

• **Bronze checks:** confirm that all six Bronze tables exist and contain records.

• **Silver reconciliation:** compare Bronze schedule records with accepted and rejected Silver schedule records. The expected relationship is Bronze schedules = accepted Silver schedules + rejected Silver schedules.

• **Gold checks:** confirm that all nine Gold tables exist and contain records.

• **KPI consistency:** compare flight and cancellation totals in fact_flight_activity with the corresponding totals in mart_flight_daily.

• **Final report:** display the validation outcomes and counts of passed and failed checks.

"This gives us a final control point. The Gold notebook creates the business-ready tables; the validation notebook checks that the pipeline has produced the expected outputs and that important totals agree."

8. Machine Learning – Flight Cancellation Prediction

"In addition to the analytics pipeline, we prepared a machine-learning notebook for flight cancellation prediction. The notebook is an experimental modelling component, separate from the main production workflow."

"It creates a feature table called **ml_flight_cancellation_features** using historical flight information. Cancellation status is converted into a target label, where 1 represents cancelled and 0 represents not cancelled."

"The notebook prepares categorical and numerical features, handles missing values, encodes categorical values and standardizes numerical features. It then splits the data chronologically, using the earlier 80 percent for training and the later 20 percent for testing."

"We train and compare two classification models: Logistic Regression and Random Forest. We evaluate them using accuracy, precision, recall, F1-score and ROC-AUC. MLflow is used to track experiment parameters, metrics and trained models."

"This notebook predicts cancellation, not departure delay. It trains and evaluates models; it does not deploy a model for live predictions. The results should be treated as experimental until their performance and data quality have been reviewed."

9. Flight Operations Command Center – Dashboard

"Once the Gold tables are available, we use them in our Databricks AI/BI dashboard, called **Flight Operations Command Center**. The dashboard turns the analytical tables into visual summaries that are easier for business users to explore."

"It has four pages and global filters for Date Range, Airport, Airline and Flight Status."

Page 1 – Executive Overview

“The first page gives a high-level summary. Its KPI cards show 11,097 total flights, 2,754 cancelled flights, 160 departed flights and a cancellation rate of approximately 24.82%, along with average departure delay. The Daily Flight Operations Trend shows activity over time; Flight Status Distribution shows the status mix; Top Origin Airports by Flight Volume compares airport volumes; and Daily Cancellation Rate shows the cancellation trend by date.”

Page 2 – Airport and Airline Performance

“The second page compares airline and airport activity. Airline Flight Volume vs Cancellations compares flight volumes and cancellations by airline. The Airline Scorecard shows total flights, cancelled flights and cancellation percentage. Airport Flight Activity shows activity by origin airport, while Movement Event Distribution summarizes events such as departed, arrived, gate change and boarding started.”

Page 3 – Fleet and Schedule

“The third page focuses on fleet and scheduled operations. The KPI cards show 146 aircraft, approximately 24.47 thousand total seat capacity, approximately 11.1 thousand scheduled flights and an average route distance of approximately 934.95 kilometres. The charts cover aircraft by manufacturer, fleet status, scheduled departures per day, scheduled flights by status, seat capacity by aircraft type and aircraft by engine type.”

Page 4 – Delays, Cancellations and Weather

“The fourth page focuses on cancellations, delay reasons and weather observations. It includes Cancelled Flights per Day, Cancellation Rate by Origin Airport, Delay Reasons by Airport, Weather Condition Distribution and Weather Observations by Airport. These visuals help users explore operational patterns across cancellations, recorded delay reasons and weather observations.”

“The dashboard can show patterns, but it does not prove that weather caused a cancellation. Correlation is not the same as causation.”

10. Data Limitations and Responsible Interpretation

“There are a few data limitations that we need to be transparent about. Although the dashboard contains 11,097 flight records, only 160 records have a recorded actual departure timestamp. The delay data also contains 159 values of exactly 15 minutes and one negative value of -778 minutes.”

“Because of this, average delay and punctuality figures should be interpreted cautiously and investigated further before being treated as reliable operational performance measures. The 15-minute punctuality threshold is a provisional assumption, not a confirmed business SLA.”

“Our weather observations cover only July 1 and July 2, 2026, so the weather analysis is limited to that period. Also, total seat capacity represents aircraft capacity, not the number of occupied seats.”

11. Dashboard Publishing and Genie

“After reviewing the dashboard, we can publish it and share it with the intended stakeholders with appropriate viewing access. The dashboard is the business-facing reporting layer, while the Gold tables remain available for deeper analysis through Databricks SQL.”

“We have also prepared a Genie space with SQL-grounded questions so users can ask questions about flight operations in natural language. Response validation is still pending, so we describe Genie as an additional exploration capability rather than a fully validated production feature.”

12. Challenges and What We Learned

Pipeline challenges

• Handling missing fields, failed type conversions and invalid timestamps during Silver cleaning.

• Resolving conflicting schedule records without arbitrarily choosing a version.

• Deduplicating records without removing valid schedule versions.

• Checking relationships between operational records and airline, aircraft and airport reference data.

• Configuring Job task dependencies and investigating storage permissions for automated File Arrival triggering.

• Reconciling record counts and KPI totals across the layers.

Dashboard challenges

• Choosing the right Gold tables and measures for each dashboard visual.

• Ensuring KPI calculations agree with the underlying fact data.

• Explaining incomplete actual departure timestamps and anomalous delay values.

• Making the dashboard easy to navigate through four pages and shared filters.

“The main lesson was that building a dashboard is not only about creating charts. We first need to understand the source data, clean it, validate the calculations and be clear about the limitations before presenting the results.”

13. Closing Summary

“To conclude, our project brings together flight schedules, movement events, weather observations and reference data. We ingest the data into Bronze, clean and validate it in Silver, and create business-ready dimensions, facts and KPIs in Gold.

We use Databricks Workflows to orchestrate the stages, a validation notebook to check completeness and consistency, and an AI/BI dashboard to communicate the results. We also explored flight cancellation prediction using Logistic Regression and Random Forest with MLflow experiment tracking.

The overall outcome is an end-to-end flight operations analytics solution, from source ingestion through data processing and validation to reporting and experimental machine learning. Thank you.”

14. Quick SME Questions and Answers

Question	Suggested answer
Is this real-time streaming?	No. Auto Loader uses Structured Streaming internally, but availableNow processes currently available files and then stops. Our pipeline is incremental and batch-oriented.
What starts the workflow?	The File Arrival trigger can start the Databricks Job when files arrive in the configured Volume location.
What does Auto Loader do?	It incrementally discovers and ingests new files, using checkpoint information to track progress.
What is the difference between the trigger and Auto Loader?	The trigger starts the Job; Auto Loader discovers and ingests the files.
Why quarantine 74 schedule rows?	They belonged to 37 flight IDs with conflicting schedules, and there was no reliable effective-time field to identify the correct version.
How do you check data loss?	We reconcile Bronze schedule counts with accepted and rejected Silver schedule counts.
What does the ML notebook predict?	Flight cancellation, not departure delay. It compares Logistic Regression and Random Forest and tracks experiments with MLflow.
Are the delay KPIs reliable?	They need further investigation because only 160 records have actual departure timestamps and the delay data contains anomalies.
Does weather cause cancellations?	The dashboard can show patterns, but it cannot establish causation.
What is Storage Blob Data Contributor?	An Azure data-plane role that permits the assigned identity to read, write and manage blob data at the assigned scope. Verify the actual IAM assignment before claiming it was the role granted.